

COLLEGE BUS TRACKER: A Real-Time GPS Based Bus Tracking and Smart Bus Pass Verification System

Project-Based Placement Readiness Programme
Training and Placement Cell

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Abstract

College buses tend to arrive earlier or later than the scheduled time occasionally due to traffic, weather, and other factors, causing student and staff commuters to arrive at the bus stop and potentially miss the bus due to uncertainty about its arrival time. Proposed is an application called College Bus Tracker that uses the bus driver's phone for real-time GPS tracking of every bus and provides commuters an easy-to-use interface to view their bus's location, expected time of arrival at their bus stop, and get notified 10 minutes prior to its arrival. If a commuter misses their bus, the application helps them track the next one that would be plying on their route. Conductors and staff can also use the application to view the validity of a commuter's bus pass in a digital format with ease. The application aims to reduce the chances of a commuter missing a bus due to their ignorance of the updated arrival time and help in removing the manual process of checking bus pass validity for college transport buses.

Keywords—GPS Tracking, Mobile Application, Firebase, Bus Pass Verification, Google Maps, Real-Time Tracking

1. PROBLEM STATEMENT

College-operated buses currently do not have a way to inform a commuter about the location of their bus or the expected arrival time at their respective bus stops.

The arrival time of the bus depends on unpredictable environmental factors like traffic and weather, and hence the commuters do not have a way to get an updated estimate of when their bus would arrive. This creates two problems:

1. Missed buses: Commuters may arrive late at a bus stop and miss the bus since it already departed or arrived earlier than expected.
2. Wasted time: Commuters may arrive early at the bus stop and have to wait for the bus, whose arrival time is unknown to them, as it might be delayed due to environmental factors.

Currently, bus pass validity checks are done manually for college transport buses, which is both time-consuming and error-prone. There currently is no simple, intuitive mobile application that uses the driver's phone for tracking and helps in removing the manual process of bus pass validity checking.

2. PROPOSED SOLUTION

Our proposed solution is to develop a two-sided mobile application that would run on the Driver's phone, who drives the college transport buses, and the Student/Staff, who commute via these buses. The application would use the real-time GPS location of the driver's phone for bus tracking purposes and would not require any additional hardware on the buses for tracking purposes.

2.1 Driver-Side Workflow

The Driver would open the Driver App and select "Start" to begin tracking for their respective bus route.

The Driver App would capture the real-time GPS location of the Driver's phone and post it on the server at regular intervals. The Driver tracking would automatically stop via "Stop" when the bus reaches the college campus.

2.2 Student/Staff-Side Workflow

Every route has a set of stops that the bus passes by. Student/Staff would view the live location of their bus on a map and an estimated time of arrival (ETA) of their respective bus at their stop. The application would send out a push notification to the Student/Staff 10 minutes before the ETA of their bus at their stop. In the case where a Student/Staff misses their bus or the bus has already left, the application would show the next bus that would be plying on their route and would help them track it similarly.

2.3 Digital Bus Pass

Every Student/Staff would have a digital bus pass card showing their pass validity. The validity of the bus pass can be viewed at a glance from the color of the card:

Green: Bus pass is valid

Yellow: Bus pass is expiring soon and needs to be renewed

Red: Bus pass is invalid/expired, and the Student/Staff cannot commute

This would help conductors and staff check the validity easily without manually checking each commuter.

3. FEATURES

Live GPS tracking of the bus using the driver's phone with no additional hardware required. Live tracking map showing the location of the bus and the expected arrival time at every stop on the route. Per-stop ETA calculated based on the current position of the bus and optionally taking into account traffic conditions and other environmental factors. Push notification sent out 10 minutes prior to the ETA of the bus at the commuter's stop.

Commuter is shown the next bus on their route in the case where they miss their bus. Automated start and stop of bus tracking with just "Start" on the Driver's side and geo-fencing on the college campus to "Stop" automatically. Digital bus pass with color-coding to indicate validity at a glance with no manual checks needed. Two-sided mobile application with separate screens for the Driver and the Student/Staff.

4. ARCHITECTURE (HIGH LEVEL)

The application would use a client-server architecture, with the following components:

Driver App: This is the component that runs on the Driver's phone and captures and posts the real-time GPS location.

Backend Server: This component receives the location of the bus, computes the ETA at every stop, hosts the routes, stops, bus details, and bus pass validity, and sends out push notifications to the commuters when their bus is approaching their stop.

Student/Staff App: This is the component that runs on the Student/Staff's phone and displays the live location of their bus and ETA at their stop. It receives the push notification from the server when their bus is approaching their stop.

The flow would be that the Driver App would post the location on the server, and the server would relay the location and ETA to every Student/Staff App instance running on a phone that has subscribed to that particular bus

5. TECHNOLOGY STACK (PROPOSED)

It is proposed to use the following technology stack while developing the application, which may be modified during the Requirement Analysis and Design phase:

Mobile Application: Flutter (multi-platform mobile application using Dart)

Backend: Node.js (REST/WebAPI) with Express.js

Real-time DB: Firebase Realtime Database/WebSockets for synchronization

DB: Firebase Firestore/MongoDB

Maps SDK/Location Services: Google Maps SDK for location tracking, geofencing, and calculating ETA

Notification Services: Firebase Cloud Messaging for sending out push notifications

Auth: Firebase Auth

6. EXPECTED OUTCOME

A working prototype of the mobile application where a driver's phone GPS location is streamed into the application, which displays it on a map with an ETA. A decrease in number of missed buses due to ignorance of updated bus arrival times by the commuters, since the application would provide push notifications 10 minutes prior to the ETA at their stop.

A decrease in wasted time for commuters by providing them an accurate ETA. A decrease in time spent by conductors and staff in checking bus pass validity automatically, with a simple color-coded validity system. A reusable application framework across multiple bus routes for college transport.

7. REFERENCES

[1] Google Developers, "Maps SDK for Android – Documentation," Google, 2026. [Online]. Available: <https://developers.google.com/maps/documentation/android-sdk>. [Accessed 07 2026].

[2] Firebase Documentation, "Cloud Firestore and Realtime Database," Google Firebase, 2026. [Online]. Available: <https://firebase.google.com/docs>. [Accessed 07 2026].

[3] Firebase Documentation, "Firebase Cloud Messaging," Google Firebase, 2026. [Online]. Available: <https://firebase.google.com/docs/cloud-messaging>. [Accessed 07 2026].

[4] Flutter Documentation, "Flutter – Build Apps for Any Screen," Google, 2026. [Online]. Available: <https://docs.flutter.dev>